

Figure 5: Configuring OAuth in Jira

There are three deployments of Bitbucket:

- Bitbucket Cloud
- Bitbucket Server
- Bitbucket Data Centre



Figure 6: Configuring branch source in the Jenkins pipeline

Configuring Bitbucket Cloud

For this tutorial, we are going to use Bitbucket Cloud.

Creating an account: Go to the Bitbucket site and click on 'Get started for free'. After that, enter your valid email ID and click on 'Create Account'. Enter your name and password and click on 'Agree and Sign Up'.

Follow the instructions to verify your email address.

Creating the repository: Just click on the 'Create repository' button. After that you will see the screen shown in Figure 1.

Enter the repository's name, click on the access level (private or public) and choose 'Version control system' from Git and Mercurial. After that, click on 'Create repository'.

Push your code to the repository: First clone the repo, and then copy your code to that repo.

Jira

Jira was designed as a bug and issue tracker. But today, Jira has evolved into a powerful work management tool for all kinds of use cases, from requirements and test case management to agile software development.

The features of Jira are:

- Scrum boards
- Kanban boards
- Roadmaps
- Agile reporting
- Connecting issues to code

How to configure Jira Software Cloud

For this tutorial, we are going to use Jira Software Cloud.

Create an account: Go to the Jira cloud site, choose 'Jira software' and click on 'Try it free'. Fill all the information and create an account.

Choose a classic template: Go to Scrum. Give the project name and click on 'Create'. After that you will see the Jira dashboard.

Create an issue: Click on 'Create issue' and give the issue name. After that, click on 'Issues and filters'.

After we have done the installations, let's go about integrating Jenkins with Bitbucket and Jira.

```
imperishable@imperishable-Vostro:~/OSFY/demo$ git fetch origin
Password for 'https://akshayithape1252@bitbucket.org':
From https://bitbucket.org/akshayithape1252/demo
 * [new branch]      DEMO-1-add-feature-1 -> origin/DEMO-1-add-feature-1
imperishable@imperishable-Vostro:~/OSFY/demo$ git checkout DEMO-1-add-feature-1
Branch 'DEMO-1-add-feature-1' set up to track remote branch 'DEMO-1-add-feature-1' from 'origin'.
Switched to a new branch 'DEMO-1-add-feature-1'
imperishable@imperishable-Vostro:~/OSFY/demo$ git status
On branch DEMO-1-add-feature-1
Your branch is up to date with 'origin/DEMO-1-add-feature-1'.

nothing to commit, working tree clean
imperishable@imperishable-Vostro:~/OSFY/demo$ git status
On branch DEMO-1-add-feature-1
Your branch is up to date with 'origin/DEMO-1-add-feature-1'.

nothing to commit, working tree clean
imperishable@imperishable-Vostro:~/OSFY/demo$ ls
dashboard.php  index.php  login.php  README.md
imperishable@imperishable-Vostro:~/OSFY/demo$ vi jenkinsfile
imperishable@imperishable-Vostro:~/OSFY/demo$ git status
On branch DEMO-1-add-feature-1
Your branch is up to date with 'origin/DEMO-1-add-feature-1'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)

    jenkinsfile

nothing added to commit but untracked files present (use "git add" to track)
imperishable@imperishable-Vostro:~/OSFY/demo$ git add jenkinsfile
imperishable@imperishable-Vostro:~/OSFY/demo$ git commit -m "jenkinsfile is added"
[DEMO-1-add-feature-1 5e30e03] jenkinsfile is added
 1 file changed, 18 insertions(+)
 create mode 100644 jenkinsfile
imperishable@imperishable-Vostro:~/OSFY/demo$ git push origin DEMO-1-add-feature-1
Counting objects: 3, done.
Delta compression using up to 4 threads.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 476 bytes | 476.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0)
remote:
remote: Create pull request for DEMO-1-add-feature-1:
remote:   https://bitbucket.org/akshayithape1252/demo/pull-requests/new?source=DEMO-1-add-feature-1
remote:
To https://bitbucket.org/akshayithape1252/demo.git
 d9761a5..5e30e03  DEMO-1-add-feature-1 -> DEMO-1-add-feature-1
imperishable@imperishable-Vostro:~/OSFY/demo$
```

Figure 7: Jenkinsfile push to Bitbucket